An Integral Sliding Mode Back-EMF Observer for Position-Sensorless Permanent Magnet Synchronous Motor Drives

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***Abstract*—The continuous reaching law is generally applied to reduce the chattering in sliding mode observer for position-sensorless permanent magnet synchronous motor (PMSM) drives. However, the robustness is accordingly reduced, and there exists output error problem. To address this problem, this paper proposed an integral sliding mode observer (ISMO). Compared with the conventional observer, it can compensate the estimated back-electromotive force (EMF) error by adding the presented integral term into continuous reaching law. The experimental results show the validity of the proposed ISMO.**

Keywords—*permanent magnet synchronous motor, integral sliding mode observer, position-sensorless*

I. Introduction

Permanent magnet synchronous motors (PMSMs) have been widely used in various industrial applications, such as servo systems, textile tools, and traction systems [1]. Position-sensorless PMSM drives has been developed rapidly over the past decade due to the advantages of low cost and high reliability [2]. The existing sensorless control methods can be divided into two categories. The first category is the high-frequency injection method based on the rotor anisotropy of PMSM [3]. The second category is the back-electromotive force (EMF) method based on the mathematical model of PMSM [4].

As a typical back-EMF method, sliding mode observer (SMO) has the advantage of strong robustness to the variation of internal parameters and external disturbances [5]. However, the chattering problem, which is the inherent defect of SMO, degrades the estimation accuracy, even causing instability. In actual applications, the switching function are usually replaced by the smooth functions to eliminate the chattering [6]. However, the system state cannot follow the sliding mode surface in this quasi-sliding mode, and this will cause estimation error. In other words, the chattering elimination is at the expense of decreasing the capability of tracking the sliding mode surface [7].

In this paper, an integral sliding mode observer (ISMO) is presented to achieve finite-time convergence of system state. The steady state error can be limited even adopting the continuous reaching law, such as saturation function. The experimental results based on a 4.4kW PMSM drive system show the validity of the proposed ISMO.

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II. Conventional Observer Design

A. Mathematic Model of PMSM

The voltage equation for the stationary coordinate system of an PMSM can be expressed as

$$\begin{bmatrix} u_\alpha \\ u_\beta \end{bmatrix} = \begin{bmatrix} R_s + pL_d & \omega_e(L_d - L_q) \\ -\omega_e(L_d - L_q) & R_s + pL_d \end{bmatrix} \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} + \begin{bmatrix} e_\alpha \\ e_\beta \end{bmatrix} \tag{1}$$

where u_α u_β are stator voltages component in $\alpha\beta$-axes; R_s is the stator resistance; L_α L_β are the $\alpha\beta$-axes inductances, ω_e is the rotor electrical rotor speed; p represents e_α and e_β are the $\alpha\beta$-axes back-EMF component, respectively.

The back-EMF component for the stationary coordinate system be represented as

$$\begin{bmatrix} e_\alpha \\ e_\beta \end{bmatrix} = \left[(L_d - L_q)(\omega_e i_d - \dot{i}_q) + \omega_e \psi_f \right] \begin{bmatrix} -\sin\theta_e \\ \cos\theta_e \end{bmatrix} \tag{2}$$

where ψ_f is the permanent magnet flux linkage. From (1), the state equation of the stator current is

$$\begin{bmatrix} \dot{i}_\alpha \\ \dot{i}_\beta \end{bmatrix} = \begin{bmatrix} -\dfrac{R_s}{L_d} & -\omega_e \dfrac{L_d - L_q}{L_d} \\ \omega_e \dfrac{L_d - L_q}{L_d} & -\dfrac{R_s}{L_d} \end{bmatrix} \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} + \frac{1}{L_d} \begin{bmatrix} u_\alpha - e_\alpha \\ u_\beta - e_\beta \end{bmatrix}. \tag{3}$$

B. Conventional back-EMF Observer

According to (3), the conventional back-EMF observer can be constructed as

$$\begin{bmatrix} \dot{\hat{i}}_\alpha \\ \dot{\hat{i}}_\beta \end{bmatrix} = \begin{bmatrix} -\dfrac{R_s}{L_d} & -\omega_e \dfrac{L_d - L_q}{L_d} \\ \omega_e \dfrac{L_d - L_q}{L_d} & -\dfrac{R_s}{L_d} \end{bmatrix} \begin{bmatrix} \hat{i}_\alpha \\ \hat{i}_\beta \end{bmatrix} + \frac{1}{L_d} \begin{bmatrix} u_\alpha - z_\alpha \\ u_\beta - z_\beta \end{bmatrix} \tag{4}$$

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where "^" represents the estimated value, z_α z_β are the artificially designed reaching law according to the stator current error.

In the conventional method, the sliding mode surface $\boldsymbol{s}(x)$ and its reaching law z_α z_β are

$$\boldsymbol{s}(x)=\begin{bmatrix} s_\alpha \\ s_\beta \end{bmatrix}=\begin{bmatrix} \mathrm{sgn}(\hat{i}_\alpha - i_\alpha) \\ \mathrm{sgn}(\hat{i}_\beta - i_\beta) \end{bmatrix} \tag{5}$$

$$\begin{bmatrix} z_\alpha \\ z_\beta \end{bmatrix}=k\begin{bmatrix} \mathrm{sgn}(\hat{i}_\alpha - i_\alpha) \\ \mathrm{sgn}(\hat{i}_\beta - i_\beta) \end{bmatrix} \tag{6}$$

where k is the gain parameter of switching function. However, the discontinuous reaching law causes serious chattering phenomenon. To overcome this problem, the saturation function is often applied. The graph of saturation function is shown in Fig. 1.

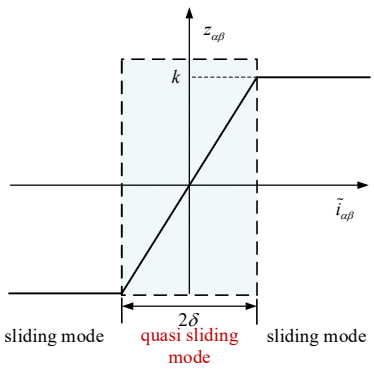

Fig. 1. Graph of saturation function

The core idea of saturation function is the combination of sliding mode and quasi sliding mode. When the current error components in α and β-axes are less than δ, the reaching law is linear one, and thus the observer no longer operates in sliding mode.

III. Proposed Integral Sliding Mode Observer

A. Estimated Stator Current Error Analysis

The estimated back-EMF component in $\alpha\beta$-axes is derived from the reaching law, which can be represented as

$$\begin{bmatrix} \hat{e}_\alpha \\ \hat{e}_\beta \end{bmatrix}=\frac{\omega_c}{s+\omega_c}\begin{bmatrix} z_\alpha \\ z_\beta \end{bmatrix} \tag{7}$$

where ω_c is the cut off frequency of low-pass filter (LPF). The effect of LPF is to filter out high-frequency harmonics. The relationship between the stator current error and reaching law in linear control law of saturation function can be described as

$$\begin{bmatrix} e_\alpha \\ e_\beta \end{bmatrix}=\begin{bmatrix} z_\alpha \\ z_\beta \end{bmatrix}=k/\delta\begin{bmatrix} \tilde{i}_\alpha \\ \tilde{i}_\beta \end{bmatrix}. \tag{8}$$

Subtracting (4) into (3)

$$\begin{aligned} \dot{\tilde{i}}_\alpha &= -\frac{R_s}{L_d}\tilde{i}_\alpha - \frac{L_d-L_q}{L_d}\omega_e\tilde{i}_\beta - \frac{1}{L_d}\tilde{e}_\alpha \\ \dot{\tilde{i}}_\beta &= -\frac{R_s}{L_d}\tilde{i}_\beta + \frac{L_d-L_q}{L_d}\omega_e\tilde{i}_\alpha - \frac{1}{L_d}\tilde{e}_\beta. \end{aligned} \tag{9}$$

Assuming that the salient polarity of PMSM can be neglected, and the Laplace transformation of (9) can be written as

$$\begin{aligned} \tilde{i}_\alpha(s) &= \frac{-1}{L_s s+R_s}\tilde{e}_\alpha(s) \\ \tilde{i}_\alpha(s) &= \frac{-1}{L_s s+R_s}\tilde{e}_\alpha(s) \end{aligned} \tag{10}$$

According (9) and (10), it can be obtained the relationship between the estimated back-EMF and that of real is

$$\begin{aligned} \hat{e}_\alpha(s) &= \frac{k/\delta}{k/\delta+L_s s+R_s}e_\alpha(s) \\ \hat{e}_\beta(s) &= \frac{k/\delta}{k/\delta+L_s s+R_s}e_\beta(s). \end{aligned} \tag{11}$$

When the PMSM drive system operates at constant speed, the relationship between k/δ and phase lag deducted from (11) is shown in Fig. 2.

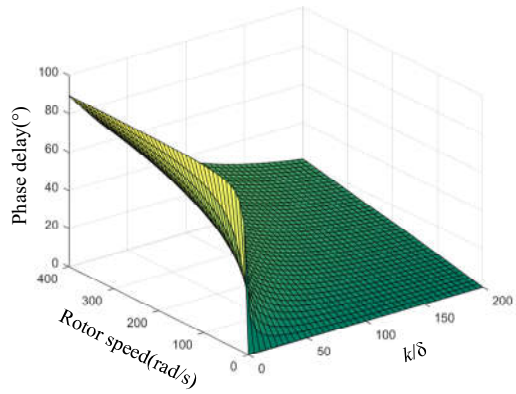

Fig. 2. The phase delay of estimated back-EMF

Though the lower slope k/δ might give rise to reduction of chattering problem, it causes phase delay in theory. Thus, there is a contradiction between chattering reduction and estimation accuracy.

B. Proposed ISMO

To solve the contradiction mentioned above, ISMO method is adopted to compensate the stator current error. The back-EMF of IPMSM can be derived from (2) that

$$\begin{bmatrix} e_\alpha \\ e_\beta \end{bmatrix}=\omega_e\psi_f\begin{bmatrix} -\sin\theta_e \\ \cos\theta_e \end{bmatrix} \tag{12}$$

It can be inferred from (12) that when the system operates at constant speed, actual back-EMF component e_α e_β are totally sinusoidal signal. According to (10) and (11), the transfer function from actual back-EMF $e_{\alpha\beta}$ to stator current error $\dot{\tilde{i}}_{\alpha\beta}$ are linear one, and that means waveforms of $\dot{\tilde{i}}_{\alpha\beta}$ are also sinusoidal.

From above analysis, the stator current error can be detected and compensated using integrator that rotates synchronously with the rotor. To achieve this, PARK and IPARK transformation is then adopted to construct the integral term, PARK and IPARK transformation can be described as

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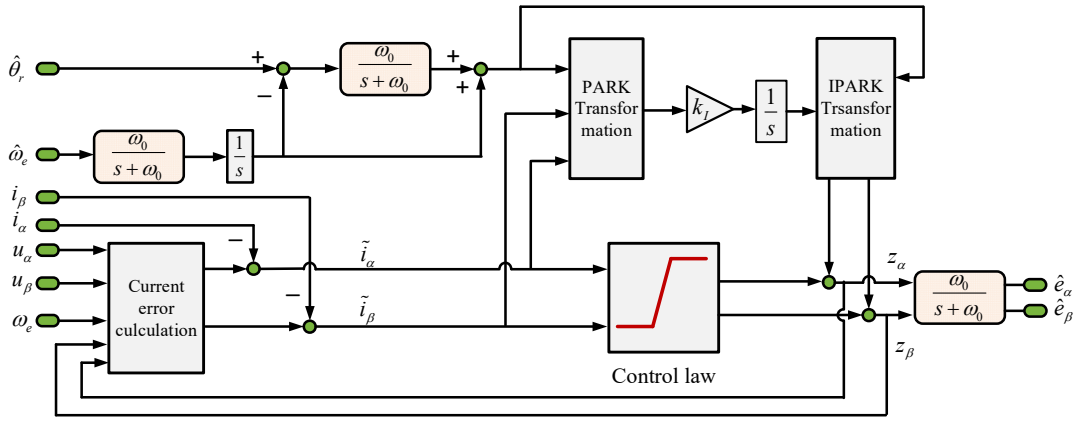

Fig. 4. Diagram of ISMO with filter

$$\begin{bmatrix} i_d \\ i_q \end{bmatrix} = \begin{bmatrix} \cos\theta_e & \sin\theta_e \\ -\sin\theta_e & \cos\theta_e \end{bmatrix} \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} \tag{13}$$

$$\begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} = \begin{bmatrix} \cos\theta_e & -\sin\theta_e \\ \sin\theta_e & \cos\theta_e \end{bmatrix} \begin{bmatrix} i_d \\ i_q \end{bmatrix} \tag{14}$$

According to (13) and (14), the integral sliding mode surface is designed as

$$s = \begin{bmatrix} \tilde{i}_\alpha \\ \tilde{i}_\beta \end{bmatrix} + k_I \begin{bmatrix} \cos\hat{\theta}\int_0^t \left(\tilde{i}_\alpha \cos\hat{\theta} + \tilde{i}_\beta \sin\hat{\theta}\right)dt - \sin\hat{\theta}\int_0^t \left(\tilde{i}_\beta \cos\hat{\theta} - \tilde{i}_\alpha \sin\hat{\theta}\right)dt \\ \sin\hat{\theta}\int_0^t \left(\tilde{i}_\alpha \cos\hat{\theta} + \tilde{i}_\beta \sin\hat{\theta}\right)dt + \cos\hat{\theta}\int_0^t \left(\tilde{i}_\beta \cos\hat{\theta} - \tilde{i}_\alpha \sin\hat{\theta}\right)dt \end{bmatrix} \tag{15}$$

where k_I represents the integral gain. the first term on the right side of equal sign is the conventional reaching law, and the second term is the proposed integral term. Then the reaching law is designed as

$$Z(s) = k\mathrm{sat}(\boldsymbol{s}(x)) \tag{16}$$

where sat( ) represents saturation function, which is used for chattering restrain. Both the chattering phenomenon and estimated back-EMF can be reduced with the proposed method, and the contradiction is solved.

C. Low-Pass Filter for Rotor Position

The proposed algorithm requires estimated rotor position, however, the error of estimated and actual rotor position may give rise to the error amplitude of estimated back-EMF. However, the conventional first order low-pass filter fails to filter out estimated rotor position without causing phase delay, thus the conventional LPF is not suit for position treatment. To solve this, a novel LPF for estimated position without phase lag is presented in (17)

$$\hat{\theta}_f = \left(\hat{\theta}_e - \hat{\omega}_e \frac{\omega_0}{s^2 + \omega_0 s}\right)\frac{\omega_0}{s+\omega_0} + \hat{\omega}_e \frac{\omega_0}{s^2 + \omega_0 s} \tag{17}$$

where $\hat{\theta}_f$ represents the output signal. When the rotor speed is constant, the calculation of steady-state error with the proposed method can be given as

$$\begin{aligned} e_{src} &= \lim_{s\to 0} sE_r(s) = e_{src} = \lim_{s\to 0} s\hat{\theta}_e(s)\left(1 - \frac{\hat{\theta}_o(s)}{\hat{\theta}_e(s)}\right) \\ &= \lim_{s\to 0} s\left(1 - \frac{\omega_c^2 + 2\omega_c s}{(s+\omega_c)^2}\right) = 0 \end{aligned} \tag{18}$$

From (18), the proposed method can filter out high-frequency components without causing phase delay. The proposed filter and the structure of ISMO with proposed LPF is shown in Fig. 3 and Fig. 4, respectively.

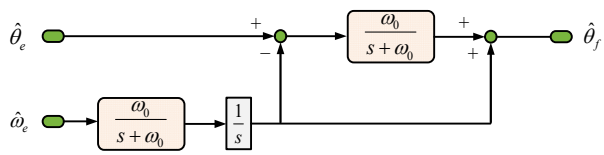

Fig. 3. The proposed low-pass filter

IV. Experimental Results

The proposed algorithm is realized on an ARM-based 4.4kW experimental platform. The calculation frequency of speed loop is set to 1 kHz, and the SVPWM switching frequency is 6 kHz. The experimental PMSM is shown in Fig.5. The parameters of the experimental PMSM are shown in Table I.

TABLE I
Parameters of the PMSM

Quantity	Value	Quantity	Value
rated power	4.4 kW	stator resistance	2.25 Ω
rated voltage	380 V	Rated torque	14.7Nm
rated current	10.3 A	*d* axis inductance	0.1544 H
rated speed	3000 r/min	*q* axis inductance	0.1544 H
pole pair number	2	Base rotor speed	100 H

Fig. 5. The photograph of 4.4kW PMSM

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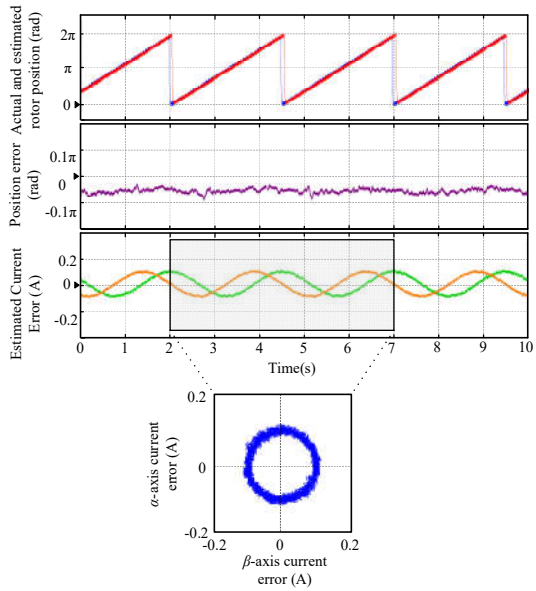
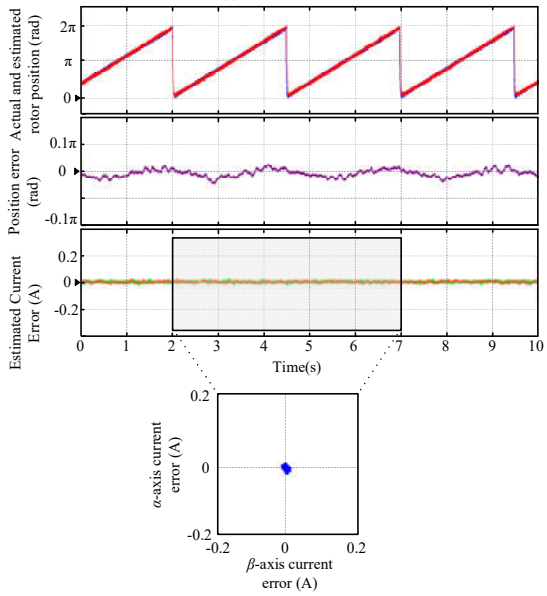
Fig. 6 shows the comparison experimental results between conventional quasi sliding mode observer and the proposed ISMO. The speed reference is set to 240rpm, and the waveforms of estimated rotor position in red, actual rotor position in blue, position error, α-axes current error in orange, and β-axes current error in green are given, respectively. Besides, the trajectory of current vector error is presented. It can be seen that the phase lag appeared in the conventional method, and the proposed method has no phase lag. In addition, the current vector error in the proposed method is much smaller.



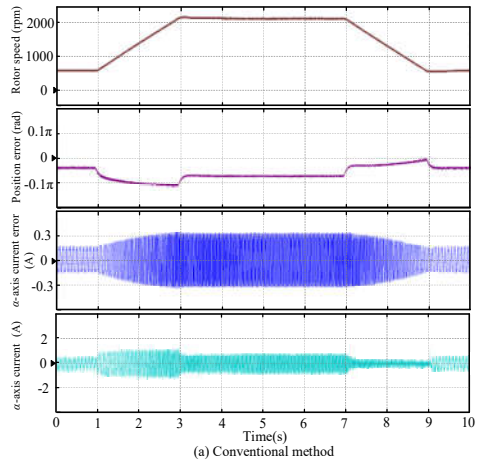
(a) Conventional method



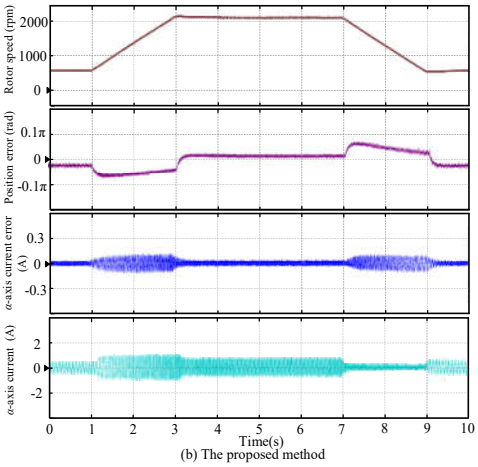
(b) The proposed method

Fig. 6. Comparison between conventional method and the proposed method with reference speed of 240rpm

To further verify the validity in full speed range, the comparison experimental results of conventional method and the proposed method is shown in Fig. 7. The waveforms of rotor speed, position error, α-axes current error, and α-axes current is presented. It can be seen that the position delay is reduced with the proposed method, and α-axes current error is greatly reduced.



(a) Conventional method



(b) The proposed method

Fig. 7. Comparison between conventional method and the proposed method with reference speed variation

From above experimental results in Fig. 6 and Fig. 7, it can be concluded that the proposed method can achieve stator current error convergence within finite time, and thus more accurate estimated rotor position can be obtained through the proposed method.

V. Conclusion

Linear control law is generally adopted to reduce chattering of SMC by establishing continuous reaching law in back-EMF observer. However, the robustness is declined and the tracking performance is decreased. This paper proposes ISMO by adding external integral term in accordance with the relationship between estimated error and back-EMF. As a result, ISMO can achieve finite-time convergence, and thus the phase delay caused by stator current estimation error can be effectively limited, the tracking performance and the robustness is then strengthened.

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